

Milan Singh Rajawat

Mechanical/Product Design Engineer

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📍 Ottawa, ON | Willing to relocate/travel

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Professional Summary

Mechanical/Product Design Engineer with 10 years of experience designing electro-mechanical systems, high-precision assemblies, and production-ready hardware across consumer, industrial and R&D environments. Skilled in Siemens NX, SolidWorks, FEA, GD&T, DFMA, tolerance analysis, and hands-on prototyping through machining and 3D printing. Experienced managing the complete product life cycle, from concept and design through validation, manufacturing transfer, and post-launch improvements. Known for strong collaboration with manufacturing, supply chain, quality, and supplier teams to deliver reliable, well-engineered products.

Core Competencies

- *Product Development:* Concept Design | New Product Development (NPD) | Product Validation | Design Reviews | Prototype Development & Testing
- *Mechanical Design:* 3D CAD Modeling | Technical Drawings | Surfacing | Rendering | Blueprints | GD&T | Engineering Calculations | Large assemblies
- *Analysis & Optimization:* Thermal/Stress Analysis | FEA | Mesh Modeling | Tolerance stack-ups/Analysis | Root Cause Analysis | Design Optimization | Cost Reduction
- *Product Lifecycle Management:* BOM Management | PLM/PDM Systems | Revision Control | Configuration Management | Technical Documentation
- *Manufacturing:* Injection Molding | Forming | Casting & Machining | Welding & Fabrication | 3D Printing | Jigs & fixture design | Mold Trials | Supplier collaboration

Work experience

Product Designer, Food Cycle Science Corp., Ottawa, ON

2023 - Present

- Spearheaded mechanical development of a commercial appliance from concept to production, delivering production-ready CAD models and detailed technical drawings using SolidWorks PDM.
- Prepared BOMs, SOPs, and ECN/ECR documentation; managed pilot runs and resolved supplier/factory issues to enable production transfer.
- Designed electro-mechanical assemblies and enclosure systems using SolidWorks, applying GD&T and DFMA principles for medium-volume production.
- Performed thermal and structural simulations in SolidWorks and validated performance under operational loads, using test data to improve design robustness.
- Reduced system heat loss by 20% through geometry optimization and targeted insulation integration validated by thermal analysis, improving overall system efficiency.
- Increased effective heat delivery to the processing chamber by 8% through a redesigned heater-bucket interface and improved thermal path.
- Designed primary user display and PCB configuration; coordinated mechanical-electrical integration and validated layout through prototype cycles.
- Developed the complete water-drainage subsystem, using key components like pump, float switches and check valve; enabling food waste throughput estimation.
- Coordinated with Asia-based suppliers and factory teams for prototyping, tooling, and production support, ensuring manufacturability and quality alignment.

Assistant Manager, Refrigerator R&D, Panasonic India Pvt. Ltd., India

2018 - 2021

- Led new product development, quality improvement, and cost-reduction projects across multiple refrigerator platforms, collaborating with manufacturing, quality, and supply chain teams to deliver robust products on schedule.
- Designed and released refrigerator components and assemblies using Siemens NX, including engineering drawings, BOMs, and engineering change implementation within a NX-Teamcenter environment.
- Redesigned refrigerator storage trays through material and structural optimization, achieving a 40% weight reduction and approximately 35% cost reduction without compromising performance.

- Reduced field-reported refrigerator dent cases by 70% through packaging redesign, transportation testing improvements, and development of localized validation standards.

R&D Engineer, Miraj Group, India

2017 - 2018

Project: Miraj Roti Master – Automated flatbread (roti) making appliance.

- Optimized electro-mechanical assemblies and mechanisms using SolidWorks; optimizing part count and manufacturability to reduce manufacturing costs by 20%.
- Developed an in-house peristaltic pump for precise liquid dosing, delivering repeatable flow control for water and oil.
- Designed compact lead-screw actuator assemblies with DC geared motors, enhancing positioning accuracy and manufacturability.
- Supported prototype testing, machine sequencing, UI validation, and troubleshooting efforts to close design gaps and improve reliability.

Design Engineer, LG Soft Pvt. Ltd., India

2014 - 2017

- Developed 3D CAD models, manufacturing drawings, and assemblies using Siemens NX for refrigerator development projects.
- Contributed to the Dual Fridge refrigerator project, helping achieve approximately 15% improvement in energy efficiency.
- Supported prototype development, first-article evaluations, and PLM documentation for production release.
- Operated and maintained in-house 3D printing equipment to accelerate prototype evaluation.

Education

- *PG Certificate in Practical Elements of Mechanical Engineering, Fanshawe College* *2022 - 2023*
(Capstone Project: Design & Development of a ball turning attachment for a Lathe machine)
- *PG Certificate in Advanced Mechanical Design, Fanshawe College* *2021 - 2022*
(Capstone Project: Design and documentation of an Automatic Tea-making machine)
- *Bachelor of Technology in Mechanical Engineering, LP University, India* *2010 - 2015*
(Capstone Project: Design & Development of Pump as a turbine system)

Technical Skills

- *CAD*: SolidWorks (advanced), PDM, Siemens NX (advanced), NX-Teamcenter, PTC Creo, AutoCAD
- *Analysis*: Thermal Analysis, Basic FEA, SolidWorks Simulation, ANSYS (Basic)
- *Quality & Design*: GD&T, FMEA, DRBFM, DFMA, TQM, APQP, DFx

Achievements & Certificates

- Professional Engineer (P.Eng.) - eligible / in progress.
- Led EB and MP production trials in China, working directly with suppliers and factory teams.
- Worked in Japan with global R&D teams, contributing to cross-regional product development.
- Earned “Expert skill badge” and “II level CAD certificate” in design by Panasonic Corp., Japan.
- Awarded with the Panasonic super star award for exceptional contribution towards growth.
- Certified in Siemens NX, Solid-Works and Auto-CAD.